

0/1 Knapsack Problem - Experimental Mode

September 17, 2026

0.1 Dynamic Programming Average Execution Time

Capacity	10	20	30	40	50	60	70	80	90	100
100	0.04051	0.05678	0.07939	0.09793	0.11858	0.14352	0.16315	0.13399	0.10983	0.12129
200	0.02405	0.04908	0.07434	0.09721	0.12069	0.14422	0.16490	0.19045	0.20815	0.23735
300	0.03461	0.07199	0.10808	0.14229	0.17497	0.21060	0.24347	0.28050	0.30967	0.32361
400	0.04401	0.08860	0.13435	0.17362	0.21726	0.26298	0.30641	0.36220	0.42591	0.47769
500	0.05883	0.12320	0.18375	0.24418	0.30081	0.36104	0.42675	0.48316	0.54442	0.62171
600	0.08021	0.15958	0.23579	0.31437	0.39194	0.45491	0.54842	0.62152	0.67034	0.73402
700	0.08284	0.17411	0.26732	0.34662	0.41012	0.50144	0.57266	0.66026	0.68021	0.79448
800	0.09009	0.17825	0.26192	0.34174	0.42864	0.51342	0.57252	0.64255	0.81585	0.81557
900	0.09153	0.18677	0.27684	0.37472	0.49442	0.64197	0.71415	0.80369	0.89779	0.97675
1000	0.11397	0.23334	0.33841	0.45261	0.55966	0.66703	0.76067	0.85409	0.91544	0.94821

0.2 Simple Greedy Average Execution Time

Capacity	10	20	30	40	50	60	70	80	90	100
100	0.00653	0.00630	0.00714	0.00768	0.00818	0.00834	0.00950	0.00733	0.00591	0.00588
200	0.00251	0.00303	0.00438	0.00393	0.00448	0.00672	0.00730	0.00817	0.00913	0.00860
300	0.00244	0.00303	0.00332	0.00370	0.00408	0.00449	0.00392	0.00410	0.00679	0.00645
400	0.00229	0.00285	0.00312	0.00349	0.00397	0.00476	0.00526	0.00599	0.00856	0.00911
500	0.00221	0.00297	0.00404	0.00479	0.00512	0.00608	0.00710	0.00688	0.00766	0.00922
600	0.00262	0.00305	0.00432	0.00503	0.00683	0.00682	0.00827	0.00959	0.00767	0.00772
700	0.00244	0.00306	0.00380	0.00442	0.00553	0.00585	0.00732	0.01005	0.00944	0.01079
800	0.00248	0.00312	0.00369	0.00402	0.00463	0.00586	0.00580	0.00601	0.00869	0.00824
900	0.00210	0.00281	0.00356	0.00341	0.00520	0.00636	0.00723	0.00513	0.00559	0.00719
1000	0.00245	0.00391	0.00385	0.00467	0.00577	0.00624	0.00695	0.00636	0.00640	0.00737

0.3 Proportional Greedy Average Execution Time

Capacity	10	20	30	40	50	60	70	80	90	100
100	0.00783	0.00856	0.01296	0.01721	0.02247	0.02802	0.03081	0.02799	0.02715	0.03126
200	0.00280	0.00452	0.00720	0.01036	0.01350	0.01673	0.01938	0.02185	0.02649	0.03175
300	0.00283	0.00454	0.00714	0.01065	0.01307	0.01584	0.01855	0.02293	0.02303	0.02423
400	0.00246	0.00384	0.00586	0.00742	0.00994	0.01248	0.01539	0.01769	0.02348	0.02659
500	0.00289	0.00503	0.00752	0.01016	0.01240	0.01636	0.02234	0.02420	0.03167	0.03498
600	0.00289	0.00452	0.00728	0.00993	0.01311	0.01634	0.02113	0.02621	0.02578	0.02980
700	0.00310	0.00425	0.00642	0.00854	0.01067	0.01395	0.01838	0.02278	0.02285	0.02726
800	0.00254	0.00402	0.00612	0.00849	0.01178	0.01630	0.01927	0.02123	0.02987	0.03099
900	0.00242	0.00432	0.00614	0.00919	0.01248	0.01741	0.02093	0.02563	0.02867	0.03240
1000	0.00261	0.00513	0.00761	0.00996	0.01384	0.01713	0.01843	0.02238	0.02381	0.02551

0.4 Simple Greedy Accuracy Compared with Dynamic Programming

Capacity	10	20	30	40	50	60	70	80	90	100
100	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
200	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
300	37.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
400	44.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
500	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
600	51.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
700	58.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
800	0.00%	27.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
900	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
1000	65.00%	51.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%

0.5 Proportional Greedy Accuracy Compared with Dynamic Programming

Capacity	10	20	30	40	50	60	70	80	90	100
100	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	33.00%	0.00%	0.00%	0.00%
200	51.00%	40.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
300	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	51.00%	0.00%	0.00%	43.00%
400	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
500	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	46.00%	0.00%	0.00%
600	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
700	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
800	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
900	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
1000	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%